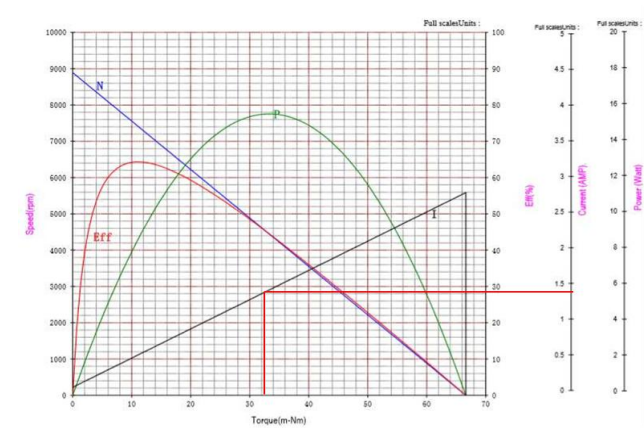
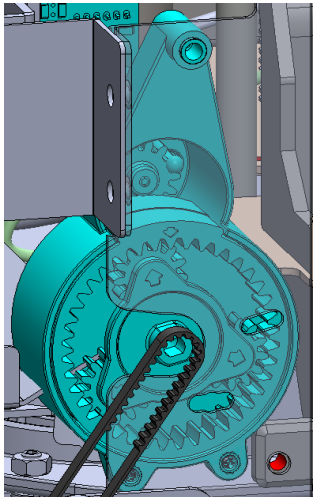


2-STAGE PLANETARY GEAR DRIVE

Input Parameters			SUMMARY OF CRITICAL POINTS		
Parameters	Value	Unit	Parameters	Value	Unit
Motor Speed	7412	rpm	Critical Bending force	7.04	kgf
Mesh Efficiency	98%	%	Output Pushing force	11.44	kgf
Input torque	0.113	kg-cm	Output Efficiency	87%	%
			Output Torque	143.12	N-mm
			Output Speed	498.2	RPM

Path	Gear #	Part Number	No. of Teeth	Module	Pitch Dia	Gear Ratio to Motor	RPM	Torque kg-cm	Torque mN-m	Tangential force (N) @ motor maximum eff. (0.55A)	Tangential force (kgf) @ motor maximum eff. (0.55A)	Tangential force (N) @ Stall (1.4A)	Tangential force (kgf) @ Stall (1.4A)	Gear Train Eff
Motor Pinion	1	Pinion	14	1	14	1.0	7412.0	0.11	11.08	1.58	0.16	4.97	0.51	1.000
Idler Gear	2	Idler	14	1	14	1.0	7412.0	0.11	10.86	1.55	0.16	4.87	0.50	0.980
Sun Gear	3	Sun Gear	14	1	14	1.0	7412.0	0.11	10.64	1.52	0.15	4.77	0.49	0.960
1st Stage carrier reduction	4	RING GEAR	40	1	40	3.9	1921.6	0.41	40.23	2.01	0.21	6.31	0.64	0.941
	5	1st stage Planet Gears	13	1	13					6.19	0.63	0.00	0.00	
2nd stage Sun Gear	6	2nd stage Sun Gear	14	1	14	3.9	1921.6	0.39	38.64	5.52	0.56	17.32	1.77	0.904
2nd stage carrier reduction	7	2nd stage Ring Gear	40	1	40	14.88	498.2	1.49	146.05	7.30	0.74	22.91	2.34	0.886
	8	2nd stage Planet Gear	13	1	13			1.46	143.12	22.02	2.24	69.09	7.04	
Belt/Pulley	9	Belt/Pulley	N/A	N/A	8	14.88	498.2	147.82	143.12	35.78	3.65	112.27	11.44	0.868



Performance (In an ambient temperature of 25-30 C)
Motor tested rapidly to prevent significant temperature rise.

At a constant voltage of	24.00	Volts
With a circuit resistance	0.000	Ohms
At No Load		
Speed :	8890	Rpm
Current:	0.111	Amp
At stall (Extrapolated)		
Torque :	66.615	m-Nm
Current:	2.793	Amp
At maximum efficiency		
Efficiency :	84.34	%
Torque :	11.071	m-Nm
Speed :	7412	Rpm
Current:	0.557	Amp
Output :	8.598	Watts
At maximum power		
Torque :	33.308	m-Nm
Speed :	4445	Rpm
Current:	1.452	Amp
Output :	15.510	Watts
Characteristics		
Torque Constant :	24.837	m-Nm/Amp
E.M.F Constant :	24.837	mV/rad/sec
D _y Resistance :	8.593	Ohms
Motor Regulation:	133.448	Rpm/m-Nm
At Torque Level:		
Torque:	m-Nm	